

IMPACT OF EXCHANGE RATE FLUCTUATIONS ON EXPORT AND IMPORT

Project Report Submitted to

CENTRE FOR BUDGET STUDIES, CUSAT

In Partial Fulfilment of the Requirement for the second semester of

**MASTER OF SCIENCE IN ECONOMETRICS AND FINANCIAL
TECHNOLOGY**

Submitted by

SIYANA YASMIN

Reg no: 24030340

Under the mentorship of

MERIN J KOMBAN

Mediassist



CENTRE FOR BUDGET STUDIES

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Kochi-682022

APRIL 2025

DECLARATION

I, SIYANA YASMIN, do hereby declare that the project titled “Impact of Exchange Rate Fluctuations on Export and Import” has been undertaken by me for the fulfilment of course outcome of second semester of Master of Science in Econometrics and Financial Technology under the mentorship of Ms. Merin J Komban, Manager, Mediassist.

I also declare that this project contains no more than an academically acceptable level of similarity and has not been submitted for the award of any degree, diploma, associateship or fellowship, or any other title in this university or any other university.

Place: Cochin

SIYANA YASMIN

Date: 30.04.2025

Reg no: 24030340

PLAGIARISM CERTIFICATE

I, SIYANA YASMIN, hereby certify that this assignment is an original effort of mine and I have not copied from anybody or from any source. The plagiarism level was found to be 1% using the iTenticate software. In case if it is found that it is not an original work of mine, then I will be liable for the same, and action can be taken against me as per the established procedures.

Date: 30.04.2025

Place: Cochin

ACKNOWLEDGEMENT

I would like to express profound gratitude to all those who contributed to the successful completion of the project.

First and foremost, I am deeply appreciative of my mentor, Ms Merin J Komban, , who supported me in the whole process and exhibits contagious enthusiasm towards the research.

I extend my sincere gratitude to Dr. Santhosh Kumar PK, Dr. Haseena Akbar, Ms. Elizabeth Joy for their constructive feedback, scholarly advice, and unwavering commitment to academic excellence.

Also, it is a matter of pleasure for me to acknowledge my deep sense of gratitude to all the faculties for their support.

I believe that this report will be an asset not only for academic institution, but will also be useful for those who are interested to learn about the project.

SIYANA YASMIN

ABSTRACT

The study explores the relationship between fluctuations in exchange rate on export and import in India, within the context of Marshall Lerner condition. Marshall Lerner condition suggests that currency depreciation improve balance of trade if the sum of the price elasticities of import and export is greater than one. The research utilizes secondary data and employs Ordinary Least Squares (OLS) regression for analysis. The findings reveal a strong positive correlation between the variables, indicating that as Indian rupee depreciates the exports and imports rise, the model only partially align with the Marshall Lerner condition. The model is statistically significant and can explain a substantial amount of the variance export and import is explained by exchange rate. This research provides valuable insights for policymakers in formulating effective strategies for maintaining balance of trade.

Key words: Exchange rate fluctuations, Export and Imports, Marshal Lerner Condition.

Table of Contents

Introduction.....	8
Theoretical Framework	9
Research Gap	9
Methodology	9
Model.....	10
Relationship Between Fluctuations in Exchange Rate on Imports and Exports.....	10
References.....	15

List of tables

Table:1 descriptive statistic	11
Table:2 correlation matrix	12
Table:3 OLS Regression (exports)	12
Table:4 OLS regression (imports)	13

List of Figure

Figure 1: Trends in exchange rate, exports and imports.....	11
---	----

Introduction

Fluctuations in exchange rates play a vital role in shaping the dynamics of a globalised economy today. The exchange rate determines the value of one currency against another, it has considerable impact on export and import of a country. This impact is due to various factors like interest rates, inflation, political events or global crisis. If a country's currency depreciates, imports become costlier, which can hurt the economy. At the same time country's exports become cheaper and more attractive. And the opposite will happen if the currency appreciates.

For developing countries like India, which rely significantly on both imports and exports for the GDP, fluctuations in exchange rate are really important. Due to massive of for crude oil, electronics, and machinery imports India depends on, the country become vulnerable to economic shocks. If the rupee depreciates, the import expense will increase and so will the fiscal and current deficit. On the other hand, IT services, pharmaceuticals, textiles, agriculture and other export driven sectors will benefit from a weaker rupee provided there is strong global demand.

However, the relationship between exchange rate movements and trade performance is not always simple or positive. Other complex interdependencies like international supply chain bottlenecks, export-import elasticity, tariff regimes, and global trade agreements also need to be factored in. Moreover, businesses involved in international trade are usually subject to foreign exchange risks that can influence profitability and pricing policies. To reduce such risks, companies can use financial instruments such as forward contracts, currency swaps, or invest in foreign subsidiaries to offset currency exposure.

This project aims to examine the impact of exchange rate volatility on a nation's import and export performance. It aims to explore how persistent fluctuations affect trade volumes, profitability, pricing, inflation, and economic health. Through the analysis of economic data, real-life case studies, and theoretical models, the study will identify challenges and opportunities brought about by currency fluctuations. In addition, it seeks to examine government policies and strategies of central banks adopted for stabilizing exchange rate fluctuations, and recommend effective business strategies for responding to such uncertainties.

Ultimately, by being aware of the far-reaching impacts of exchange rate fluctuations, stakeholders' policymakers, business executives, as well as consumers can make informed choices that yield a robust and competitive trading environment.

Theoretical Framework

Fluctuations in exchange rate significantly influence a country's exports and imports. At its core, the Marshall Lerner Condition suggests that a depreciation of domestic currency will only improve a country's balance of trade if the sum of the price elasticities is greater than one. For instance, when the rupee depreciates, Indian exports become cheaper to other countries, and imports to India become expensive. If foreign buyers increase their purchases of Indian products (exports) and Indian consumers decrease their imports because of higher prices, then the balance of trade will increase.

However, if demand for exports and imports is inelastic which means people don't change how much they buy even price change then a depreciating currency might not help the trade balance and could even make it worse. This theory plays a vital role in a country like India where exchange rates have major impact on imports and exports. The Marshall Lerner condition shows why and how individuals and companies react to price fluctuations in the global market is just as important as the currency's value.

The Marshall Lerner condition helps determine whether a currency devaluation will be a useful strategy for reducing trade deficit or increasing trade surplus. Devaluation can lead to a positive impact on trade balance if the requirement is satisfied. Devaluation, however, may actually make the trade balance worse if the requirement is not fulfilled.

Objectives

- Analyse the changes in rupee's value affect India's exports and imports

Research Gap

While numerous studies examined the impact of exchange rate on international trade, there exist gap in how it will affect developing economies like India. The majority of current research concentrates on aggregated data and long-term trends, thereby ignoring short term volatility and sectoral specific implication that are crucial for successful policy responses.

Methodology

The methodology used for this study includes quantitative approach which uses secondary data from reserve bank of India (RBI) and world bank. The study covers the period from 2000 to 2022. In this study, the dependent variables are exports and imports, and independent variable is exchange rates. Ordinary least square(OLS)

regression and descriptive statistics are used as analytical methods to find patterns and relationship in the dataset.

Model

With the theory of international trade and Marshall Lerner condition as the theoretical foundation the project goal is to investigate the relationship between exchange rate(the independent variable) fluctuations and export and import(the dependent variable). To fulfil the study objective the following model have been constructed.

Export=f (exchange rate)

Econometric model:

Exports:

$$LEXP = \alpha + \beta(LEXC) + \varepsilon$$

LEXP= Exports in log form

LEXC=Exchange rate in log form

Imports:

$$LIMP = \alpha + \beta(LEXC) + \varepsilon$$

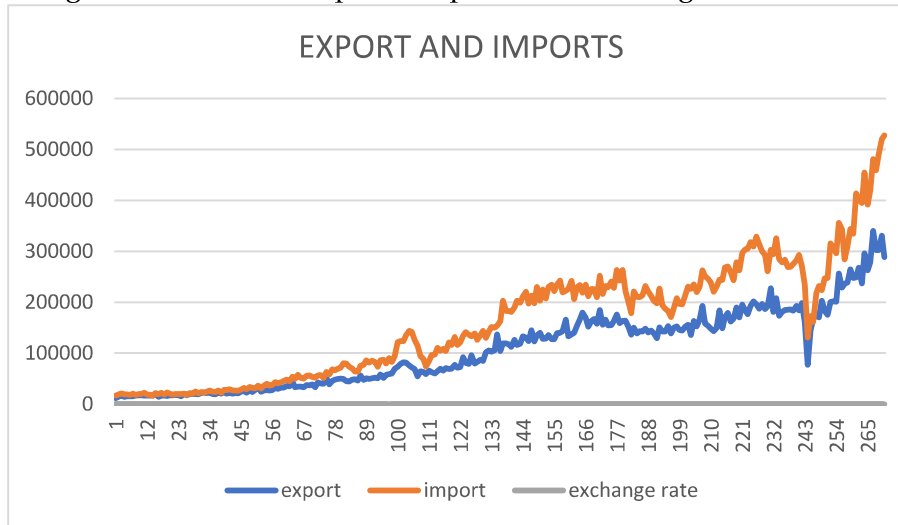
LIMP=imports in log form

LEXC=exchange rate in log form

Relationship Between Fluctuations in Exchange Rate on Imports and Exports

An analysis of the of the association between exchange rate on export and import within the framework of Marshall Lerner condition using log transformed data, the analysis revealed insightful findings.

Figure 1: Trends in imports, exports and exchange rate



Source: Author's calculation

Figure 1 illustrates the trends in export, import and exchange rates from 2000 to 2022. Over this period, both exports and imports show an upward trend, with imports exceeding exports, which indicates trade deficit. In the early years the exchange rate is relatively stable. It began to rise in around 2010 and reflecting currency depreciation. in 2020 there is a considerable downfall in all three indicators due to covid-19. post 2020 there is recovery in three variables.

Table:1 descriptive statistic

	LEXPORT	LIMPORT	LEXC
Mean	11.27143	11.62676	3.991933
Median	11.5675	11.94582	3.891136
Std. Dev.	0.88669	0.963078	0.202544
Skewness	-0.476083	0.602156	0.324312
Kurtosis	1.895968	2.040689	1.585669
Jarque-Bera	24.00055	26.76855	27.33764
Probability	0.000006	0.000002	0.000001
Observations	271	271	271

Source: Author's calculation

The variability of LEXPORT is around 0.8866 and of LIMPORT is 0.9631 indicating import has slightly higher variability in the data. LEXPORT has negative skewness of -0.4761 which indicates a slight left tail. While LIMPORT and LEXC have positive skewness. Which means both are skewed to right. All the three variables is platykurtic distributions because kurtosis value is less than 3. The jarque bera and lower p values indicates that none of the variables is normally distributed.

Table:2 correlation matrix

	LEXPORT	LIMPORT	LEXC
LEXPORT	1	0.991826149	0.780940988
LIMPORT	0.991826149	1	0.73725503
LEXC	0.780940988	0.73725503	1

Source: Author's calculation

The table 2 analyses the correlation between the variables such as export import and exchange rate on log form. The correlation coefficient indicates the strong positive correlation between variables which indicate the increase in exchange rate will leads to increase in exports and imports. Both exports and imports have strong positive correlations with exchange rate. This indicate that changes in exchange rate impact both trade activities and they tend to rise together.

Table:3 OLS Regression (exports)

Variable	Coefficient	standard error	t-Statistic	probability
C	-2.376077	0.666378	-3.565661	0.0004
LEXC	3.418772	0.166717	20.50638	0.0000
R-squared			0.609869	
Adjusted R-squared			0.608419	
F-statistic			420.5117	
Prob(F-statistic)			0.000000	

Source: Author's calculation

The table 3 indicates the Ordinary Least Square Regression results to determine the relationship between the variables. The model as per the result is shown below.

$$LEXPORT = -2.376077 + 3.418772(LEXC) + \epsilon$$

The t statistic indicates the significance of individual parameters. At a 5% level of significance, the coefficient is significant. The coefficient of LEXC is positive and significant. Which means that a 1% increase in exchange rate is associated with 3.42% increase in exports on an average.

The F-statistic indicates that the model is statistically significant and the given data is fit to the model. The R-squared value indicates that around 61% of variation in export is explained by the changes in exchange rate. The adjusted R -square also supports the goodness of fit. The results indicate strong positive relationship between the variables.

Table:4 OLS regression (imports)

Variable	Coefficient	standard error	t-Statistic	probability
C	-2.367255	0.782896	-3.02372	0.0027
LEXC	3.505575	0.195868	17.8976	0.0000
R-squared		0.543545		
Adjusted R-squared		0.541848		
F-statistic		320.3242		
Prob(F-statistic)		0.00000		

Source: Author's calculation

Table:4 indicates the ordinary least square regression results to determine the relationship between the variables.

$$\text{LIMPORT} = -2.367255 + 3.505575(\text{LEXC}) + \varepsilon$$

The t-statistic indicates the significance of individual parameters. At a 5% level of significance, the coefficient is significant. The coefficient of LEXC is positive and significant. Which means that a 1% increase in exchange rate is associated with 3.51% increase in imports on an average.

The F-statistic indicate that the model is statistically significant. The r square value indicates that around 54.35% of the variation in imports is explained by changes in exchange rate. The adjusted r square also supports the goodness of fit of the model. The overall results show strong positive relationship between exchange rate and imports

Conclusion

The study investigated the relationship between fluctuations in exchange rate on export and import of India from period 2000-2022 thereby exploring the applicability of Marshall Lerner condition in this context. Making use of log transformed data the analysis revealed insightful findings. The regression analysis using OLS revealed that when the rupee depreciates both exports and imports rise. This means that the trade is sensitive to exchange rate. When rupee depreciates, Indian goods become cheaper for foreign buyers, which increases exports. At the same time, imported goods become more expensive. But the demand for imported goods is high so imports continue to grow.

In Indian context the Marshall Lerner condition is partially true , because both the imports and exports increases with depreciation .however since India depends heavily on imported goods including essential items a depreciating rupee means higher import costs which increase country's trade deficit and a pressure on economy. This means that a weaker rupee can help Indian exporters but it negatively

affects consumers and business who rely on imported goods. Many export industries in India rely on imports for their raw materials so thereby increasing their costs also.

Overall, the study shows that exchange rates matter a lot for trade in India. But managing them to is not easy task. The government and central bank need to analyse other variables not just exchange rate. India needs appropriate policies to increase exports and reduce imports and manage inflation. And businesses should be vigilant and prepared for facing risk associated with currency fluctuations and use tools like forward contracts or currency hedging to protect themselves.

References

- **Reserve Bank of India (RBI).** (2000–2022). *Handbook of Statistics on Indian Economy*. Retrieved from <https://www.rbi.org.in>
- **World Bank.** (2023). *World Development Indicators*. Retrieved from <https://databank.worldbank.org>
- Krugman, P. R., & Obstfeld, M. (2009). *International Economics: Theory and Policy* (8th ed.). Pearson Education.
- Dornbusch, R. (1976). Expectations and Exchange Rate Dynamics. *Journal of Political Economy*, 84(6), 1161–1176. <https://doi.org/10.1086/260506>
- Bahmani-Oskooee, M., & Niroomand, F. (1998). Long-run price elasticities and the Marshall–Lerner condition revisited. *Economics Letters*, 61(1), 101–109. [https://doi.org/10.1016/S0165-1765\(98\)00151-8](https://doi.org/10.1016/S0165-1765(98)00151-8)